

SIH26054 — 6-Slide PPT Content Blueprint

AI-Enabled Real-Time Digital Twin for MALE UAV Aero Piston Engines

(Judge’s-eye structuring — what wins points at each slide)

SLIDE 1 — Literature Review & Research Gap

Judge’s lens: This slide answers “Do they actually understand the field, or are they just building a dashboard?” A strong Slide 1 makes the panel *feel* the gap before you’ve even proposed your idea.

Slide title suggestion: “Why Existing Approaches Fail Military-Grade MALE UAV Engines”

Layout: A 4-row comparison table (or 4 stacked cards) — one row per paradigm — ending in a punch-line strip that transitions into your idea.

#	Existing Paradigm	Strength	Fatal Flaw (for this use case)
1	Pure Data-Driven RUL (LSTM/Transformer on C-MAPSS-style data)	Captures complex non-linear sensor correlations without needing engine thermodynamics	Purely statistical — under unseen conditions (e.g. sudden high-altitude pressure drop) it can output thermodynamically impossible predictions
2	Traditional Physics-Informed Neural Networks (PINNs)	High interpretability, strictly obeys thermal/structural laws	Rigid & compute-heavy online; solves forward equations well but can’t do active, multi-objective, closed-loop decisions
3	Standard Deep RL for control	Excellent sequential decision-making, adaptive optimization	Black-box exploration — trial-and-error behavior is unacceptable in a live military UAV engine
4	Conventional Modular Digital Twins (isolated state-estimation / anomaly / RUL blocks)	Clean, debuggable software engineering	Silent failure propagation — a null/delayed payload from one module is silently treated as “nominal” downstream, fabricating

Optional citation strip below the table (keep to 3-4, short form): - Deep-learning RUL benchmarks (MDPI/IEEE, C-MAPSS-based LSTM/Attention networks) - Hybrid Kalman-Filter + MLP digital twins for aircraft engines - Online-adaptive PINN frameworks - IEEE Transactions work on digital twins for UAV-assisted edge environments

Closing line of the slide (the “so what”): *“No single paradigm gives you physical trustworthiness AND adaptive decision-making AND fault-tolerant orchestration at once. Our architecture is the first to combine all three.”*

Diagram idea: A simple 2×2 or funnel graphic — four boxes (Data-driven / PINN / DRL / Modular DT) each with a small ✕ icon next to their flaw, converging with an arrow into a single box labelled “Our Solution: Agentic PINN + DRL” on the bottom-right. This visual alone tells the whole slide’s story at a glance — good if you want minimal text.

SLIDE 2 — Proposed Solution Overview

Judge’s lens: This is where they decide “is this idea coherent and does it actually solve the stated problem?” Be concrete about what it is, not just what it uses.

Slide title suggestion: “Our Solution: Agentic Physics-Informed Digital Twin”

One-line pitch (put this large, at the top): > A real-time digital twin that fuses Physics-Informed Neural Networks, Deep Reinforcement Learning, and a LangGraph-orchestrated AI agent to predict, explain, and safely act on aero piston engine health — with zero tolerance for silent failure.

What problem it solves (map directly back to Slide 1 gaps): - Eliminates physically-impossible predictions → PINN enforces thermodynamic law inside the loss function - Eliminates unsafe black-box control actions → DRL action space is boundary-constrained by the PINN - Eliminates silent data-cascade failures → LangGraph agent actively audits and halts on malformed/delayed sensor payloads

Core Features (good as 4-6 icon+text tiles): 1. **Real-time telemetry ingestion** — RPM, CHT, EGT, oil pressure, fuel flow, vibration, MAP, AFR 2. **Physics-constrained anomaly & stress estimation** (PINN layer) 3. **Adaptive prognostics** — continuous Remaining Useful Life (RUL) estimation via DRL 4. **Self-auditing agent loop** — sensor-health cross-checking before any prediction is trusted 5. **3D interactive digital twin dashboard** — exploded engine view with live thermal heatmaps 6. **Mission-reliability decision support** — helps ground stations plan recovery/abort maneuvers before failure

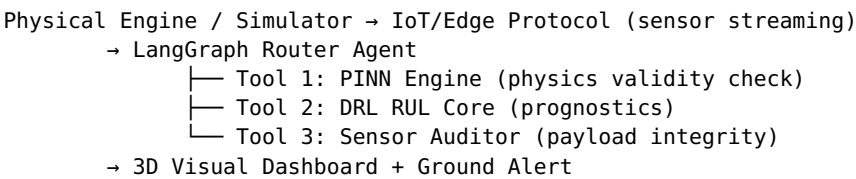
Diagram idea for this slide: A “solution overview” graphic — center circle labelled “LangGraph Orchestrator,” three surrounding satellite nodes (PINN / DRL / Sensor Auditor), and an outer ring showing inputs (engine sensors) flowing in and outputs (3D dashboard + ground alerts) flowing out. This is essentially a simplified version of your agent-loop diagram from the PDF, redrawn cleanly for a general audience — keep labels minimal, this slide shouldn’t look “technical” yet (save deep technical detail for Slide 3).

SLIDE 3 — Technical Approach, Architecture & Math

Judge’s lens: This is the technical-rigor slide. Judges will scan for: is the architecture end-to-end coherent, is the math real (not decorative), and is the tech stack sensible for the claimed performance. Don’t overload — pick the strongest 3 elements.

Slide title suggestion: “Technical Architecture: PINN-DRL-Agentic Pipeline”

A. System Workflow (put as a horizontal flow diagram)



B. The Three Functional Layers (short bullets, one per layer)

- PINN Layer — “Physics Guardian”:** continuous estimator of structural/thermal stress; forces predictions to obey Fourier’s law of heat conduction and gas-law transformations.
- Deep RL Layer — “Prognostic Strategist”:**
 - State Space: sensor streams + PINN’s current physical gradient
 - Action Space: throttle/fuel-air correction recommendations + RUL trend projection
 - Reward Function: rewards flight-duration extension, heavily penalizes actions pushing components toward wear thresholds
- LangGraph Agent Loop — “Orchestrator”:** cyclically routes between the three tools instead of a one-way pipeline, so it can catch and halt anomalies mid-process rather than passing them downstream.

C. Key Mathematics (use exactly 1-2 equations — don’t crowd the slide)

Loss function (PINN):

$$L_{\text{Total}} = L_{\text{Data}} + L_{\text{Physics}}$$

Physics constraint term (Fourier’s law based):

$$L_{\text{Physics}} = (1/N) \sum | \partial T / \partial t - \alpha \nabla^2 T - \dot{q}_{\text{combustion}} |^2$$

One-line explainer under the equation: “This term penalizes any predicted temperature trajectory that violates real heat-transfer physics — so the network literally cannot output an impossible reading.”

D. Tech Stack (compact table or icon row)

Layer	Tools
AI/ML Modeling	Python, PyTorch/TensorFlow (PINN + DRL)
Orchestration	LangGraph agentic framework

Backend	FastAPI, WebSockets
Visualization	Three.js / React Three Fiber / WebGL (custom fragment shaders for heatmaps)
Data	NASA C-MAPSS-style datasets, IEEE Dataport profiles for engine telemetry simulation

Diagram idea for this slide: Use the workflow block diagram above as the *main visual* — it’s the single most important image in your deck since it proves the architecture is real and closed-loop (not just a linear pipeline). Optionally pair it with a small inset of the 3D dashboard mockup (exploded engine, red/yellow/green component health) to show the output side. Keep the math as a clean formula card, not squeezed text.

SLIDE 4 — Feasibility & Viability

Judge’s lens: Judges specifically probe: can this be built in the hackathon/PoC timeframe, and does it scale to a real deployment? Be honest and structured — feasibility slides that oversell get torn apart in Q&A.

Slide title suggestion: “Feasibility & Viability”

Technical Feasibility: - Built entirely on open-source, well-supported frameworks (PyTorch/TensorFlow, FastAPI, LangGraph, Three.js) — no proprietary dependency risk - PINN and DRL are established research techniques individually; novelty is in the *orchestrated combination*, which reduces “unproven technology” risk - Public datasets (NASA C-MAPSS-style, IEEE Dataport) allow model development/validation without needing live UAV hardware access at prototype stage - Digital twin visualization can be demonstrated on a simulator before hardware/telemetry integration

Operational Feasibility (DRDO context): - Directly addresses DRDO’s stated need: near-zero downtime for military-grade MALE UAV engines - Agent-based auditing (Tool 3: Sensor Auditor) is specifically designed to prevent the exact failure mode DRDO would care about most — a system silently reporting “nominal” when it isn’t - Architecture is modular — PINN/DRL/Agent tools can be upgraded independently without redesigning the whole pipeline

Economic/Resource Viability: - Open-source stack keeps licensing cost near zero - Compute cost is the primary overhead (PINN + DRL training/inference) — manageable at prototype/demo scale, and edge-inference optimization is a clear future-work item - Reduces long-term maintenance cost vs. traditional modular DT pipelines by catching failures early (avoids costly field failures / unscheduled maintenance)

Risks & Mitigation (a judge will ask about this — pre-empt it): | Risk | Mitigation | |—|—| | PINN training instability on real noisy telemetry | Start with simulator-validated data, progressively introduce noise | | DRL reward mis-specification leading to unsafe suggestions | Hard physics-based action boundary from PINN (already part of design) | | Latency of 3D visualization under real-time load | WebSocket streaming + shader-based rendering keeps compute off the render thread |

Diagram idea: Not essential here — a simple feasibility “traffic-light” checklist (Technical / Operational / Economic, each marked green) reads faster than a diagram and keeps the slide readable.

SLIDE 5 — Impact & Benefits

Judge's lens: This is the “why should DRDO care” slide — tie every point back to mission reliability, not generic AI buzz.

Slide title suggestion: “Impact: From Reactive Maintenance to Predictive Mission Assurance”

Direct Impact: - Prevents mid-flight engine failure by catching anomalies before they cascade — directly supports mission reliability for MALE UAVs - Gives ground stations actionable RUL data to make real-time go/no-go and recovery-maneuver decisions - Removes “silent failure” risk that plagues conventional modular digital twins — a core safety improvement, not just a performance one

Strategic / Defence Impact: - Reduces unplanned downtime → higher UAV fleet availability for missions - Physics-constrained predictions build trust for use in safety-critical, high-stakes defence systems (unlike black-box-only ML approaches) - Extensible to other DRDO aero-propulsion systems beyond piston engines (turboprop, hybrid-electric UAV powertrains) as future scope

Illustrative Benefit Framing (qualitative — present as directional claims, not as validated field results): - Fewer physically-impossible/false predictions vs. pure data-driven models, because of the embedded physics constraint - Faster anomaly detection vs. static modular pipelines, because the agent audits continuously instead of at fixed checkpoints - Improved failure transparency — every halted/flagged prediction has a traceable reason (sensor fault vs. genuine anomaly), aiding maintenance crews

Diagram idea: A simple “before vs. after” split panel — left side “Traditional Modular Pipeline” with a broken/dead-end arrow at a failure point, right side “Our Agentic Digital Twin” with the same failure point caught and rerouted. This visually sells the safety story without needing hard numbers you haven't field-validated.

Judge's caution for you: Avoid presenting the specific percentage figures (e.g., “~70% error reduction,” “5000% scalability gain”) as measured results unless you actually benchmark them — judges will ask “how did you get this number?” If you don't have a real experiment behind it, frame these as *target design goals* instead of achieved metrics. This is the single most common way strong-looking SIH decks lose credibility in Q&A.

SLIDE 6 — References

Judge's lens: A clean references slide signals rigor. Keep it to your 4 core literature comparisons plus any dataset/framework references — formatted consistently.

Suggested layout: Small font, numbered list, two columns if needed.

1. Deep learning (LSTM/Attention) approaches to Remaining Useful Life prediction using C-MAPSS-style aero-propulsion prognostics benchmarks (MDPI/IEEE shared benchmarks).
2. Hybrid digital twin approaches combining Kalman filtering with multi-layer perceptrons for electric/piston aircraft engines.
3. Online-adaptive Physics-Informed Neural Network frameworks for remaining-useful-life prediction.
4. Digital twin frameworks for UAV-assisted edge environments (IEEE Transactions), covering modular health/sensing/trajectory pipelines.

5. NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) dataset — reference dataset class for engine degradation simulation.
6. IEEE Dataport — supplementary aero-engine sensor data profiles.

Note: fill in exact author names, publication years, and full citation format (IEEE style) before submission — the PDF you shared has placeholder-style citations, so double check each against the actual paper before presenting them as verified sources.

Quick Deck-Level Advice

- **Diagram density:** Slides 1, 2, 3 are your visual-heavy slides. Slides 4–6 should be mostly clean text/tables — don't force a diagram where a checklist or table communicates faster.
- **Consistency:** Reuse the same three icons/colors for PINN (physics/blue), DRL (strategy/green), Agent (orchestration/orange) across Slides 2, 3, and 5 so judges visually “lock in” your architecture's identity across the deck.
- **Q&A prep:** Be ready to defend (a) why LangGraph specifically vs. a simpler state machine, and (b) what happens if the PINN and DRL disagree — these are the two questions a technically sharp judge will ask first.